

Lab 2
Programing Lab With Python
BCACC393

1. Write a Python program to create a string and display its length using the `len()` function.
2. Write a Python program to accept two strings from the user and concatenate them.
3. Write a Python program to accept a string and display it in **uppercase and lowercase**.
4. Write a Python program to accept a string and count the number of occurrences of a particular character using `count()`.
5. Write a Python program to accept a string and replace one word with another using the `replace()` method.
6. Write a Python program to accept a student's name and age and display the information using an **f-string**.
7. Write a Python program to accept the first name and last name and display the full name using an **f-string**.
8. Write a Python program to create a list of five numbers and display its length, maximum value, and minimum value.
9. Write a Python program to create a list of five numbers and calculate the sum of all elements using `sum()`.
10. Write a Python program to create a list and add an element at the end using `append()`.
11. Write a Python program to create a list and insert an element at a specified position using `insert()`.
12. Write a Python program to create a list and remove a specified element using `remove()`.
13. Write a Python program to create a list and remove the last element using `pop()`.
14. Write a Python program to create a list of numbers and sort the elements in ascending order using `sort()`.
15. Write a Python program to create a tuple of five numbers and display its length, maximum value, and minimum value.
16. Write a Python program to create a tuple and count how many times a particular element occurs using `count()`.
17. Write a Python program to create a tuple and find the position of a particular element using `index()`.
18. Write a Python program to convert a list into a tuple and display both the original list and the resulting tuple.
19. Write a Python program to accept a student's name, roll number, and marks and display all details using an **f-string**.
20. Write a Python program to accept the name and three subject marks of a student, calculate the total and average, and display the result using **f-strings**.
21. Write a Python program to create a nested list containing three lists of numbers and display all the elements of the nested list.
22. Write a Python program to create a nested list representing marks of three students in three subjects and calculate the total marks of each student.
23. Write a Python program to create a nested list and access a specific element using row and column indexing.
24. Write a Python program to create a nested tuple containing the names and marks of three students and display the details of each student.
25. Write a Python program to create a nested tuple of numbers and calculate the sum of the elements of each inner tuple.

26. Write a Python program to create a nested list of numbers and find the maximum value from each inner list.
27. Write a Python program to create a nested list containing two lists of numbers and add the corresponding elements of the two lists.
28. Write a Python program to create a nested tuple containing student names, roll numbers, and marks. Display each student's details using indexing and an f-string.

Learning Objectives

After completing this laboratory, students will be able to:

1. **Understand and apply fundamental Python programming concepts**, including variables, data types, input/output, operators, type casting, and basic built-in functions.
2. **Perform basic operations on strings, lists, and tuples** using methods and functions such as len(), count(), replace(), append(), insert(), remove(), pop(), sort(), index(), sum(), min(), and max().
3. **Use f-strings for formatted output** and develop simple Python programs involving user input, calculations, and presentation of results.
4. **Create and manipulate nested lists and tuples**, access elements using indexing, and perform basic operations on multidimensional data structures.